



MATHS SPECIAL BATCH

Algebra Sheet-5

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1. If $x + y = 41$, then find $(x - 20)^{2021} + (y - 21)^{2021}$?

यदि $x + y = 41$ है, तो $(x - 20)^{2021} + (y - 21)^{2021}$ ज्ञात कीजिये?

A) 0

B) 1

C) 2

D) 3

2. Simplify $(5z - 7y)^2 + (7z + 5y)^2 - 49z^2$

$(5z - 7y)^2 + (7z + 5y)^2 - 49z^2$ को सरल कीजिए (SSC CGL MAINS 2024)

A) $25z^2 + 74y^2$

B) $35z^2 + 70y^2$

C) $26z^2 + 82y^2$

D) $30z^2 + 83y^2$

3. If $x = a + \frac{1}{a}$ and $y = a - \frac{1}{a}$ then $\sqrt{x^4 + y^4 - 2x^2y^2}$ is equal to :

if $x = a + \frac{1}{a}$ and $y = a - \frac{1}{a}$ then $\sqrt{x^4 + y^4 - 2x^2y^2}$ is equal to :

(a) $16a^2$ (b) 8 (c) $8/a^2$ (d) 4

4. If $x+y=2z$, then $\frac{x}{x-z} + \frac{y}{y-z} = ?$

a) 0 b) 1 c) 2 d) 3

5. If $\alpha + \beta + \gamma = 0$, then find $\frac{3\beta^2 + \alpha^2 + \gamma^2}{2\beta^2 - \alpha\gamma}$?

यदि $\alpha + \beta + \gamma = 0$, तो $\frac{3\beta^2 + \alpha^2 + \gamma^2}{2\beta^2 - \alpha\gamma}$ ज्ञात करें?

SSC CPO 2024

A) $\frac{1}{2}$

B) 1

C) 2

D) $1/4$

6. If $a+b+c = 0$, then the value of

$\frac{2a^2}{(b^2+c^2-a^2)} + \frac{2b^2}{(a^2+c^2-b^2)} + \frac{2c^2}{(a^2+b^2-c^2)}$ is equal to:

(a) 3 (b) -3

(c) -4 (d) 0

7. If $a+b+c=2s$, then $\frac{(s-a)^2 + (s-b)^2 + (s-c)^2 + s^2}{a^2 + b^2 + c^2} = ?$

a) $a^2 + b^2 + c^2$ b) 0 c) 1 d) 2

8. If $2s=a+b+c$, then what is $s(s-a)(s-b)(s-c) \left[\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} - \frac{1}{s} \right]$ equal to?

यदि $2s=a+b+c$, तो $s(s-a)(s-b)(s-c) \left[\frac{1}{s-a} + \frac{1}{s-b} + \frac{1}{s-c} - \frac{1}{s} \right]$ किसके बराबर है?

(UPSC CDS-1 2025)

A) abc

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- B) $2abc$
 C) $4abc$
 D) $ab+bc+ca$

9. If $pq+qr+rp=0$, then $\left(\frac{p^2}{p^2-qr} + \frac{q^2}{q^2-rp} + \frac{r^2}{r^2-pq}\right) = ?$

- a) 3 b) 1 c) 2 d) 0

10. If $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = 1$, then find $\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} = ?$

- a) 0 b) -2 c) 3 d) 1

11. If $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 0$, then $\frac{ac}{b^2} + \frac{b^2}{ac} - \frac{c^3}{a^3} = ?$

- a) -3 b) -2 c) -1 d) 0

12. If $a + b + c = 1$ and $a^3 + b^3 + c^3 = 4$, then $\frac{1}{a+bc} + \frac{1}{b+ac} + \frac{1}{c+ab} = ?$

- a) 1 b) -1 c) -2 d) 3

13. If $\left[\sqrt{(a^2 + b^2 + ab)}\right] + \left[\sqrt{(a^2 + b^2 - ab)}\right] = 1$, then what is the value of $(1-a^2)(1-b^2)$?

यदि $\left[\sqrt{(a^2 + b^2 + ab)}\right] + \left[\sqrt{(a^2 + b^2 - ab)}\right] = 1$ है, तो $(1-a^2)(1-b^2)$ का मान क्या है?

- (a) $1/4$ (b) $4/7$
 (c) $5/4$ (d) $3/4$

14. What is the value of $\frac{a^2+ac}{a^2c-c^3} - \frac{a^2-c^2}{a^2c+2ac^2+c^3} - \frac{2c}{a^2-c^2} + \frac{3}{a+c}$?

$\frac{a^2+ac}{a^2c-c^3} - \frac{a^2-c^2}{a^2c+2ac^2+c^3} - \frac{2c}{a^2-c^2} + \frac{3}{a+c}$ का मान क्या है?

- (A) 0 (B) 1 (C) $\frac{ac}{a^2+c^2}$ (D) $\frac{6}{a+c}$

15. What is $\frac{x^2-y^2-z^2-2yz}{x^2+y^2-z^2+2xy} + \frac{x^2-y^2-z^2-2yz}{x^2-y^2+z^2-2xz}$ equal to?

$\frac{x^2-y^2-z^2-2yz}{x^2+y^2-z^2+2xy} + \frac{x^2-y^2-z^2-2yz}{x^2-y^2+z^2-2xz}$ किसके बराबर है? (CDS 2023)

- A) $\frac{x}{x+y-z}$
 B) $\frac{y+z}{x+y-z}$
 C) $\frac{2x}{x+y-z}$
 D) $\frac{2y+2z}{x+y-z}$

16. What is $\frac{1}{x(x-y)(x-z)} + \frac{1}{y(y-z)(y-x)} + \frac{1}{z(z-x)(z-y)}$ equal to?

$\frac{1}{x(x-y)(x-z)} + \frac{1}{y(y-z)(y-x)} + \frac{1}{z(z-x)(z-y)}$ किसके बराबर है?

- A) 0
 B) 1
 C) $1/xyz$
 D) $-1/xyz$



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17. If $a+b+c=8$ and $ab+bc+ca=5$, then $bc(b+c)+ca(c+a)+ab(a+b)+3abc=?$

- a) 13 b) 40 c) 50 d) 32

18. If $ab + bc + ca = 119$, $a^2 + b^2 + c^2 = 162$ and a, b & c are positive values then what is the value of $a^2(b+c)+b^2(c+a)+c^2(a+b)+3abc$?

यदि $ab + bc + ca = 119$, $a^2 + b^2 + c^2 = 162$ और a, b और c धनात्मक हैं तो $a^2(b+c)+b^2(c+a)+c^2(a+b)+3abc$ का मान क्या है?

- (a) 2380 (b) 2499 (c) 2450 (d) 1760

19. What is the simplified value of $\frac{(x+y+z)(xy+yz+zx)-xyz}{(x+y)(y+z)(z+x)}$?

$\frac{(x+y+z)(xy+yz+zx)-xyz}{(x+y)(y+z)(z+x)}$ का सरलीकृत मान क्या है? (Mains 2021)

- A) z
B) x
C) 1
D) y

20. If $\frac{m-a^2}{b^2+c^2} + \frac{m-b^2}{c^2+a^2} + \frac{m-c^2}{a^2+b^2} = 3$, then find the value of m ?

- a) $a^2 + b^2 - c^2$ b) $a^2 + b^2$
c) $a^2 + b^2 + c^2$ d) $a^2 - b^2 - c^2$

21. If $\frac{x-a^2}{b+c} + \frac{x-b^2}{c+a} + \frac{x-c^2}{a+b} = 4(a+b+c)$, then find the value of x ?

- a) $(a+b+c)^2$ b) $a^2 + b^2 + c^2$
c) $ab+bc+ca$ d) $a^2 + b^2 + c^2 - ab - bc - ca$

22. If $a^2 = by + cz$, $b^2 = cz + ax$, $c^2 = ax + by$, then $\frac{1}{2}[\frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z}] = ?$

- a) 1 b) $\frac{1}{2}$ c) $a+b+c$ d) 2

23. If $b+c=ax$, $c+a=by$, $a+b=cz$, then the value of $\frac{1}{9}[\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}]$ is:

यदि $b+c=ax$, $c+a=by$, $a+b=cz$ है तो $\frac{1}{9}[\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}]$ का मान क्या है?

- a) 0 b) 1 c) $\frac{1}{9}$ d) $\frac{1}{3}$

24. If $P = \frac{x^4-8x}{x^3-x^2-2x}$, $Q = \frac{x^2+2x+1}{x^2-4x-5}$ and $R = \frac{2x^2+4x+8}{x-5}$, then $(P \times Q) \div R$ is equal to:

यदि $P = \frac{x^4-8x}{x^3-x^2-2x}$, $Q = \frac{x^2+2x+1}{x^2-4x-5}$ and $R = \frac{2x^2+4x+8}{x-5}$ है, तो $(P \times Q) \div R$ किसके बराबर है ?

- a) $\frac{1}{2}$ b) 1 c) 2 d) 4

25. If $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$, then the value of $c + \frac{1}{a}$ is

यदि $a + \frac{1}{b} = 1$ और $b + \frac{1}{c} = 1$ हो तो $c + \frac{1}{a}$ का मान क्या होगा?

- (a) 0 (b) 2 (c) 1 (d) 3



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26. If $\frac{A}{L} + \frac{M}{B} = 1$ & $\frac{B}{M} + \frac{N}{C} = 1$, Then find $\frac{L}{A} + \frac{C}{N}$?

यदि $\frac{A}{L} + \frac{M}{B} = 1$ & $\frac{B}{M} + \frac{N}{C} = 1$, तो $\frac{L}{A} + \frac{C}{N}$ ज्ञात कीजिए?

- A) 1
- B) B/M
- C) M/B
- D) 0

27. If $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$ (where $a \neq b \neq c$), then abc is equal to
(Mains 2016)

यदि $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$ (जहाँ $a \neq b \neq c$) तो abc किसके बराबर है?

- (a) +1
- (b) -1
- (c) +1 & -1
- (d) None of these/ इनमें से कोई नहीं

28. The factors of $(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$ are (Mains 2014)

$(a^2 + 4b^2 + 4b - 4ab - 2a - 8)$ गुणनखंड क्या होंगे?

- (a) $(a - 2b - 4)(a - 2b + 2)$
- (b) $(a - b + 2)(a - 4b - 4)$
- (c) $(a + 2b - 4)(a + 2b + 4)$
- (d) $(a + 2b - 1)(a - 2b + 1)$

29. If $(a+b+c)=20$ & $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 30$, then find the value of $(\frac{a}{b} + \frac{b}{a} + \frac{b}{c} + \frac{c}{b} + \frac{c}{a} + \frac{a}{c})$?

यदि $(a+b+c)=20$ & $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 30$ है, तो $(\frac{a}{b} + \frac{b}{a} + \frac{b}{c} + \frac{c}{b} + \frac{c}{a} + \frac{a}{c})$ का मान ज्ञात कीजिये?

- A) 597
- B) 600
- C) 599
- D) cannot find

30. If $ax+by=1$ & $bx+ay=\frac{2ab}{a^2+b^2}$, then the value of x (in terms of a and b) is?

यदि $ax+by=1$ & $bx+ay=\frac{2ab}{a^2+b^2}$, तो x का मान (a और b के संदर्भ में) क्या है?

- A) $\frac{2b}{a^2+b^2}$
- B) $\frac{a}{a^2+b^2}$
- C) $\frac{b}{a^2+b^2}$
- D) $\frac{2a}{a^2+b^2}$